How the flux sector was determined on the (J_x, J_y) plane

A methods record for the cubic-16 XYZ scan

5 August 2026

Abstract

This note records exactly how the (J_x, J_y) phase diagrams were computed and how the 0-flux / π-flux label was assigned, including the two things that make the answer non-obvious: the label cannot be read off the raw cluster, and the winding-free label needs no counterterm. The headline numbers: the raw ground state reports positive hexagon flux in all 1681 cells of the plane, whereas the winding-free (masked) ground multiplet gives a flux that is quantised to $-1/4$, $+1/4$ or exactly zero, and is nonzero *if and only if* the protected ground multiplet is six-fold degenerate. Every validation, every failure and every caveat is listed.

1 Model, parametrisation, and what was scanned

The cubic-16 cluster of the pyrochlore lattice carries 16 sites, 8 tetrahedra, 36 four-loops and 16 contractible hexagons, with 90 two-in–two-out (ice) configurations. With $J_z \equiv J_{zz} = 1$ the Hamiltonian at zero twist is linear in the two transverse couplings,

$$H \;=\; \underbrace{\tfrac{1}{2}\sum_t \big(n_t^{\uparrow} - 2\big)^2}_{\text{Ising}} \;-\; J_{\pm}\sum_{\langle ij\rangle}\big(S_i^+ S_j^- + \text{h.c.}\big) \;+\; J_{\pm\pm}\sum_{\langle ij\rangle}\big(S_i^+ S_j^+ + \text{h.c.}\big), \tag{1}$$

so the three pieces are assembled once and recombined per grid point. The scan is over the square $J_x, J_y \in [-1, 1]$ at $J_z = 1$ on a 41×41 grid, with the dominant-a mapping

$$J_{\pm} = -\frac{J_x + J_y}{4}, \qquad J_{\pm\pm} = \frac{J_x - J_y}{4}. \tag{2}$$

The XXZ line of the earlier campaign is the diagonal $J_x = J_y$; the 56-cell $(J_{\pm}, J_{\pm\pm})$ campaign grid occupies a narrow diagonal band, drawn as open circles on every panel.

Because $J_{\pm\pm} \neq 0$ breaks S^z conservation, the working basis is the full $2^{16} = 65\,536$ spin basis, not a fixed-magnetisation block.

2 The naive phase diagram

Everything in ***`fig_xy_phase`*** *and* ***`fig_xy_flux`*** *is ordinary exact diagonalisation*: full spectrum, no ice projection, no mask, no counterterm. Per grid point the ten lowest levels are obtained with `eigsh` and the ground state gives the ice weight $Z = \langle\psi|P_{\text{ice}}|\psi\rangle$, the mean ice violation $\langle\sum_t Q_t^2\rangle$, the gap, and the two loop amplitudes of Sec. 3.

A solver note that mattered. The first pass stalled: one point consumed 8174 s and four array tasks died on a three-hour wall or on `ArpackNoConvergence`. The cause was not iteration count but Krylov space — `eigsh(k=10)` defaults to `ncv`$= 21$, far too tight where the low levels are clustered below 10^{-4}. Setting `ncv`$= 80$ (escalating to 180 and 300), seeding with the ice-uniform vector and warm-starting from the previous point reduced the same points to 6–11 s, a factor above 800 on the worst one. The 861-point upper triangle then completed in under ten minutes.

Missing cells. Of 164 cells lost to the failed tasks, 74 were recovered by the reflection of Sec. 6, 82 from the per-point lines the dead tasks had already printed, and the remaining 8 by re-solving; the final assembly uses no scraped values at all. The internal symmetry check returns 1.5×10^{-13} on the gap and 1.7×10^{-12} on the ice violation. It returns 2.9×10^{-3} on the ice weight, which is not bookkeeping error but the genuine eigenvector ambiguity in the near-degenerate region; it is left visible rather than removed by rebuilding the plane from one triangle.

3 The flux observable

The 0-flux / π-flux label of a quantum spin ice is the sign of the contractible hexagon amplitude. For a hexagon h with sites ordered around the ring,

$$O_{\rm hex}^{(h)} \;=\; S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^- \;+\; \text{h.c.}, \qquad O_{\rm hex} \;=\; \frac{1}{N_{\rm hex}} \sum_h O_{\rm hex}^{(h)}, \tag{3}$$

with $N_{\rm hex} = 16$ the contractible hexagons (those of zero winding). In the S^z product basis $O_{\rm hex}^{(h)}$ is a permutation-like operator: it is nonzero only on the two configurations in which spins alternate around the ring, and it maps each to the other by flipping all six. Including both alternating patterns is what makes it Hermitian. The same construction on the 36 wrap-around four-loops defines O_4, the finite-size artefact amplitude. Positive $\langle O_{\rm hex} \rangle$ is 0-flux, negative is π-flux.

3.1 Why the raw cluster gets the sign wrong

Measured on the raw ground state, $\langle O_{\rm hex} \rangle$ is *positive in all* 1681 *cells*, ranging only over $+0.013$ to $+0.123$: raw ED labels the entire plane 0-flux and never produces a π-flux signature anywhere. That is an artefact, and its mechanism is the one this project exists to remove. The wrap-around amplitude $g_4 = 4J_\pm^2/J_{zz}$ is **even** in $J_\pm$ while the physical ring exchange $g_6 = 12|J_\pm|^3/J_{zz}^2$ is **odd**. For $J_\pm > 0$ the two reinforce and the raw sign is accidentally right; for $J_\pm < 0$ — every cerium pyrochlore — they oppose, and on this cluster the artefact wins. Table 1 shows the inversion directly, with $\langle O_4 \rangle$ sitting 3–6× above $\langle O_{\rm hex} \rangle$ in the raw state and collapsing once the winding is removed.

3.2 Attribution: the operator is right, the state was wrong

Before accepting that reading, the discrepancy was localised. The raw full-basis measurement was reproduced with the *reference* code path — the $S^z = 0$ translation-sector basis and the counterterm campaign's own `loop_expectation` — on the bare Hamiltonian. At $J_\pm = -0.10$ both give $E = -0.502949$, gap 0.10603, $Z = 0.7346$, $\langle O_{\rm hex} \rangle = +0.0914$, $\langle O_4 \rangle = +0.3264$, agreeing to every printed digit. The Hamiltonian conventions were checked to match as well (`amplitude`$= -J_\pm \times$phase in both). The entire gap to the cached -0.209 is therefore H versus $H + C$: a difference of state, not a bug in the observable.

	naive ED			winding-free $H+C$		
$J_\pm$	Z	$\langle O_{\rm hex}\rangle$	$\langle O_4\rangle$	Z	$\langle O_{\rm hex}\rangle$	$\langle O_4\rangle$
−0.02	0.962	+0.115	0.356	0.986	−0.248	0.022
−0.05	0.888	+0.107	0.348	0.907	−0.238	0.056
−0.10	0.735	+0.091	0.326	0.692	−0.209	0.056
−0.15	0.613	+0.079	0.307	0.477	−0.177	0.058
−0.20	0.525	+0.069	0.290	0.326	−0.149	0.066
−0.25	0.461	+0.063	0.278	0.230	−0.122	0.092
−0.30	0.415	+0.058	0.269	0.172	−0.067	0.147
−0.35	0.379	+0.054	0.261	0.144	−0.009	0.178
+0.046	0.723	+0.119	0.335	0.752	+0.215	0.193

Table 1: The XXZ line. Naive ED reports 0-flux at every coupling; the winding-free state is π-flux throughout $J_\pm < 0$, with the magnitude decaying as the band dissolves. Naive ED also overstates the ice weight once $|J_\pm| > 0.1$, because the artefact binds the ice manifold more tightly than the physics does.

4 The winding-free determination

4.1 The mask, and what it is applied to

Splitting the Hilbert space with $P = P_{\rm ice}$ (rank 90) and $Q = 1 - P$ (rank 65 446) and eliminating Q exactly gives the Feshbach map

$$F(z) \;=\; PHP \;+\; \underbrace{PHQ\,(z - QHQ)^{-1}QHP}_{\Sigma(z)}, \tag{4}$$

whose self-consistent roots $E = \operatorname{eig} F(E)$ are exact eigenvalues of H with nonzero ice weight. **The mask deletes every entry of $F(z)$ connecting different transport (polarisation) sectors.** It is one operation, `sector_project` in `DownfoldBlocks.f_matrix`.

It is worth being explicit about what the mask is *not*, because the natural guess is wrong. It is not the deletion of cross-sector matrix elements of H: the ice block PHP is diagonal, so there are no such elements to delete. Every winding amplitude is generated by excursions through the spinon space and therefore lives entirely in the self-energy $\Sigma(z)$. Deleting entries of H would accomplish nothing; deleting them from the fold is the whole method.

4.2 Why no counterterm is involved

The counterterm $C = -\big(\text{transport-changing part of } F(z_0)\big)$, supported inside the ice manifold, is the operator that makes $H + C$ a genuine Hamiltonian on the full space whose fold equals the masked map at the anchor z_0. It exists because the masked map returns energies and ice-block vectors but no states above the band and no full-space wavefunctions — and it is exact only at z_0, which is the origin of the residual z-dependence of order $1 - Z$.

None of that is needed here. A contractible hexagon flip takes an ice configuration to an ice configuration, so $O_{\rm hex}$ closes inside the 90-dimensional manifold and is a dense 90×90 matrix in the ice basis. The masked map's own ground multiplet is therefore sufficient, and the flux reported below is computed with the mask alone.

4.3 Extraction, step by step

At each grid point:

1. build H from Eq. (1) on the full 65 536 basis at the $(J_\pm, J_{\pm\pm})$ of Eq. (2);
2. block-diagonalise and fold, obtaining PHP, the poles $\{\varepsilon_q\}$ of QHQ and the rotated coupling (Sec. 5);
3. solve $E = \operatorname{eig}\Delta[F(E)]$ for all 90 branches by guaranteed bisection, where Δ is the sector mask. A branch with no root below the lowest pole has migrated into the continuum and is counted as lost — that count is the band-completeness diagnostic, not an error;
4. take the lowest root E_0 and the multiplet of branches within 10^{-9} of it; diagonalise $\Delta[F(E_0)]$ and keep those eigenvectors $\{\phi_m\}$;
5. report $\langle O_{\text{hex}}\rangle = \frac{1}{M}\sum_m \phi_m^\dagger O_{\text{hex}}\phi_m$ and the ice weight $Z = \frac{1}{M}\sum_m \big[1 - \phi_m^\dagger\,\partial_z\Sigma(E_0)\,\phi_m\big]^{-1}$.

Averaging over the multiplet is mandatory. The masked ground state is degenerate by theorem, not by accident: the sector structure protects a multiplicity of six at weak coupling and two after it reorganises. A single eigenvector of a degenerate subspace is whatever LAPACK returned, so every expectation value is averaged over the multiplet. The spread across multiplet members is recorded and is below 10^{-6} (typically 10^{-31}) at *all* 379 points with a surviving band, so the averaging changes no value while removing the basis dependence.

5 Numerics

The one expensive step is the eigendecomposition of QHQ. On the full basis that is a single dense solve of dimension 65 446, which not only costs ~ 9 h but *segfaults*: $n^2 = 4.3\times10^9$ exceeds 2^{31} and overflows LAPACK's `int32` indexing. It is not a memory problem and no amount of RAM fixes it.

5.1 The Klein group

The pyrochlore cubic cell contains four FCC primitive cells, so four translations map the 16-site cluster to itself:

$$t_0 = (0,0,0),\quad t_1 = \big(0,\tfrac12,\tfrac12\big),\quad t_2 = \big(\tfrac12,0,\tfrac12\big),\quad t_3 = \big(\tfrac12,\tfrac12,0\big),\tag{5}$$

in units of the cubic edge. Each doubles to a cubic lattice vector, so $t_i^2 = \mathbb{1}$ on the cluster, and the product of any two distinct non-identity translations is the third, $t_1t_2 = t_3$. The group is therefore $\mathbb{Z}_2\times\mathbb{Z}_2$ — the Klein four-group V_4, not the cyclic $\mathbb{Z}_4$ — and this is what the "Klein" in `build_blocks_klein` refers to. All four irreps are one-dimensional with characters ±1, labelled $++$, $+-$, $-+$, $--$.

Two facts make it usable. Both terms of Eq. (1) are translation invariant, so all four elements commute with H at *any* $(J_\pm, J_{\pm\pm})$ — the blocking survives switching on the pair term, which is what breaks S^z and forces the full basis in the first place. And the ice manifold is closed under them, so the P/Q split commutes with the decomposition. Hence QHQ block-diagonalises into four blocks of $\approx 16\,384$: the flop count falls by $\sim 16\times$ (four solves of $n/4$ against one of n), peak memory from ~ 100 GB to ~ 10 GB, and each block has $n^2 = 2.7\times10^8$, comfortably inside `int32`.

Within each irrep k the ice components are $I_k = B_k[\text{ice}]^\dagger$, of size $m_k\times 90$; an SVD gives an orthonormal O_k of rank d_k with $\sum_k d_k = 90$. Projecting the ice subspace out, $(1 - O_kO_k^\dagger)H_k(1 - O_kO_k^\dagger)$, leaves d_k *spurious* ice-kernel modes alongside the genuine QHQ poles. These must be separated by their overlap with O_k and **not** by eigenvalue, since genuine poles cross zero at strong coupling and an eigenvalue threshold would discard real physics.

That classification is precisely what fails at the four unresolved points below: the overlaps stop being bimodal, so no threshold separates 36 genuine from the 38–47 modes that look spurious. The Klein decomposition itself is exact; it is the heuristic that sorts its output which breaks.

Cost: 174 s per point on an H100 against 2600 s on 16 CPU cores, so the production run used GPUs — 861 points over 20 array tasks, 3 h 52 m wall. Results are checkpointed after every point and the script resumes from disk, so a wall-clock overrun costs one point rather than a task.

Four points did not resolve. The ice-kernel classification failed at $(J_x, J_y) = (-1, +1)$, $(-0.75, +1)$, $(0, +1)$ and $(+1, +1)$, with between 38 and 47 modes looking spurious against an SVD rank of 36 and overlap ranges reaching from $\sim 10^{-31}$ (or exactly 0) to 1. All four sit on the $J_y = +1$ boundary at maximal coupling, where extra degeneracy makes the overlaps ambiguous; the failure is a limitation of the classification heuristic, not of the mask. They are recorded as unresolved and left blank.

5.2 Symmetries not yet exploited

The Klein decomposition is the only reduction currently implemented, and it is not the only one available. All of the following are exact at any $(J_\pm, J_{\pm\pm})$ unless stated.

$U(1)$ (XXZ line only). At $J_{\pm\pm} = 0$ total S^z is conserved and the problem lives in the 12 870-state $S^z = 0$ block, which is why the counterterm campaign was inexpensive along the line. The pair term destroys it, so this buys nothing off the diagonal.

$\mathbb{Z}_2$ parity (everywhere), worth $8\times$. The pair term shifts $N_\uparrow$ by ± 2 and the exchange preserves it, so $\Pi = (-1)^{N_\uparrow}$ commutes with H for all couplings. Every ice state has $N_\uparrow = 8$ exactly — each site belongs to exactly one A-tetrahedron, so summing the ice rule over the four A-tetrahedra gives $4 \times 2 = 8$ — hence P lies entirely in the even sector, and $\Sigma(z) = PHQ(z - QHQ)^{-1}QHP$ cannot reach the odd sector at any order. The 32 768 odd states are therefore not a block to be diagonalised separately but irrelevant, and can be dropped: QHQ falls from 65 446 to 32 678, the Klein blocks from $\approx 16\,362$ to $\approx 8\,170$.

reduction	blocks	dimension	relative flops	memory/block
none (direct)	1	65 446	—	*int32 overflow*
Klein V_4 (implemented)	4	16 362	1	2.14 GB
+ parity	4	8 170	1/8	534 MB
+ global spin flip	8	4 085	1/32	134 MB

Table 2: Available reductions of QHQ. Parity is exact and nearly free to implement: restrict the basis before building anything and the existing Klein path runs unchanged. At 174 s per point, the parity factor alone would bring the full plane from ~ 42 to ~ 6 GPU-hours — and would have made the abandoned CPU route viable at ~ 325 s per point.

Global spin flip (everywhere), worth a further $4\times$. $X = \prod_i \sigma_i^x$ commutes with all three terms of Eq. (1), preserves the ice manifold and $N_\uparrow$ parity, and leaves O_{hex} + h.c. invariant. It is more invasive than parity: it *permutes* the polarisation sectors ($p \to -p$) rather than fixing them, so it interacts with the mask's sector labelling and with the SVD-based ice-kernel separation — the one step that already fails at four points. Worth taking, but as a separately validated exercise, not as part of the same change.

Validation any such change must pass. Re-solve points that already exist and require agreement to machine precision. The flux is quantised at $\pm 1/4$ and so is insensitive to small errors; the ice weight and the band gap are the discriminating checks.

6 The exchange symmetry, and its verification

$J_x \leftrightarrow J_y$ sends $J_{\pm\pm} \to -J_{\pm\pm}$ at fixed $J_\pm$ and is an exact symmetry, implemented by the diagonal transformation $D = i^{N_\uparrow}$: the pair term picks up $i^{\pm 2} = -1$ while the exchange term, which conserves $N_\uparrow$, is untouched. The hexagon operator has three S^+ and three S^-, hence $i^3(-i)^3 = 1$, and is invariant. Only the upper triangle therefore needs solving: 861 points cover all 1681 cells.

This was verified rather than assumed, and the check had to be designed with some care — the assembled figure writes both (i, j) and (j, i) from one solve, so a symmetry check computed from it returns zero by construction and means nothing. Instead both members of six mirror pairs (with $J_{\pm\pm} = \pm 0.25$, genuinely different matrices) were solved independently. They agree *bitwise* in energy, gap, ice weight, ice violation, hexagon flux and winding amplitude — as expected when the symmetry acts by a diagonal similarity, since Lanczos from the same real seed then builds an identical tridiagonal matrix.

7 Results

7.1 Three phases, and a quantised label

region	upper triangle	full plane	ground multiplet
π-flux, $\langle O_{\text{hex}}\rangle = -1/4$	101	193	six-fold
0-flux, $\langle O_{\text{hex}}\rangle = +1/4$	16	29	six-fold
no ring resonance, $\langle O_{\text{hex}}\rangle = 0$	262	497	one-, two- (or 90-)fold
no ice-descended band	478	962	—
unresolved	4	7	—
total	861	1681	

Table 3: Winding-free classification of the plane. Only 13% of the plane carries a flux at all, and the overwhelming majority of that is π — the label naive ED never produces anywhere.

Two features are sharper than expected. First, the flux is **quantised**: among all resonant cells the magnitude takes the single value $|\langle O_{\text{hex}}\rangle| = 0.25$, with the individual numbers equal to $\pm 1/4$ to 10^{-16} and nothing in between. Second, the resonant set coincides *exactly*, cell by cell, with the set where the protected ground multiplet is six-fold: all 117 six-fold cells are resonant, and all 262 cells of multiplicity 1, 2 or 90 have zero flux. The protected multiplet is what carries the flux; where the protection is lost, the resonance is gone with it.

A value of exactly 1/4 is what a uniform-amplitude resonance over the ice manifold would give: with the normalisation of Eq. (3), $\langle O_{\text{hex}}\rangle$ for an equal-amplitude state equals $\langle n_{\text{flip}}\rangle / N_{\text{hex}}$, so 1/4 corresponds to a mean of exactly four flippable hexagons per ice configuration. That reading is consistent with the measurement but has not been verified here, and is offered as an interpretation only.

7.2 Naive against winding-free

The two diagrams disagree about the plane at the coarsest level, not in detail. Naive ED returns a confident ice weight, gap and positive flux in every one of the 1681 cells. The winding-free map says that 57% of the plane has no ice-descended band at all, a further 30% has a band carrying no hexagon resonance, and the 13% that does carry a flux is mostly π. The disagreement over the *existence* of the band is arguably the more important of the two.

The winding-free diagram also reproduces, by an independent route, the dissolution threshold found in the $J_{\pm\pm}$ study: the campaign cell $(J_\pm, J_{\pm\pm}) = (-0.05, 0.30)$ maps to $(J_x, J_y) = (0.7, -0.5)$ and lands in the no-band region, consistent with the band dying once $J_{\pm\pm}/|J_\pm| \gtrsim 5$. A substantial number of the 56 campaign cells lie in the no-resonance region rather than in the π-flux disc.

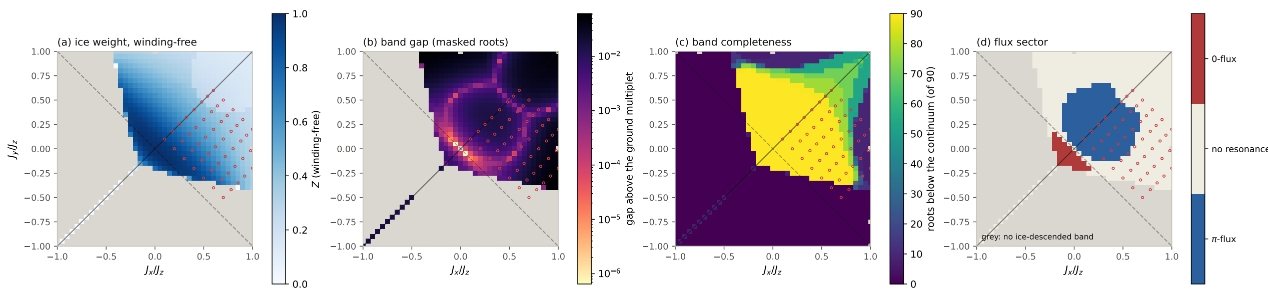

Figure 1: The winding-free plane. (a) ice weight of the masked ground multiplet; (b) gap above the protected multiplet *within the ice-descended band* — not the same object as the naive full-spectrum gap, and not to be compared with it; (c) band completeness, roots surviving out of 90; (d) flux sector, categorical. Grey is “no ice-descended band”. The ring of near-degeneracies in (b), encircling the origin at roughly the boundary of the π-flux disc, is an observation from the data and has not been identified here.

8 What is not established

- **The naive ice-violation panel has no winding-free counterpart.** $\langle \sum_t Q_t^2 \rangle$ requires the full-space wavefunction, which is exactly what the masked map does not carry and what the counterterm exists to supply. Band completeness occupies that panel instead.
- **Panel (b) is a different quantity** from the naive gap panel: the gap above the protected multiplet inside the band, versus the gap of the full spectrum. They are labelled separately and should not be overlaid.
- **The masked flux is an ice-manifold expectation**, taken on the normalised 90-dimensional vector, whereas the cached $H + C$ numbers are full-state expectations; the two are related by $\langle\psi|O_{\text{hex}}|\psi\rangle = Z\langle\phi|O_{\text{hex}}|\phi\rangle + (Q \text{ terms})$. At $J_\pm = -0.10$, $0.692 \times 0.25 = 0.173$ against a cached -0.209, leaving -0.036 from the spinon dressing; at $+0.046$, $0.752 \times 0.25 = 0.188$ against $+0.215$. Signs agree and residuals are of the expected size, but the two conventions are not interchangeable.
- **Four boundary points are unresolved** and are not extrapolated over.
- **The $1/4$ interpretation** of the preceding section is a conjecture, not a derivation.

- **One cluster.** Everything here is cubic-16; nothing in this note addresses the thermodynamic limit, and the cluster is the error bar.

9 Reproduction

file	role
`campaign/scan_xy_phase_plane.py`	naive ED over the plane, one task per column
`campaign/scan_xy_points.py`	naive ED at a point list, with the `ncv` ladder
`campaign/fill_xy_phase_gaps.py`	reflection, log recovery, assembly of the naive plane
`campaign/scan_xy_masked_flux.py`	masked Feshbach map, flux and ice weight
`campaign/check_flux_symmetry.py`	mirror-pair test of $J_x \leftrightarrow J_y$
`campaign/make_xy_phase_figure.py`	naive plane, three panels
`campaign/make_xy_flux_figure.py`	naive flux, artefact, and the XXZ inversion
`campaign/make_xy_masked_figure.py`	winding-free plane, four panels
`campaign/make_naive_vs_wf_figure.py`	the XXZ-line contrast of Table 1
`src/qsi_campaign/downfold.py`	`build_blocks_klein`, `solve_roots`, the mask

Production jobs: naive plane `18332139` (plus the repair chain), naive flux `18341289`/`18341649`, mirror check `18341657`, winding-free plane `18365005`. Raw records are `campaign/outputs/xy_phase_plane/{xyscan,flu` the masked records carry both the masked and unmasked fold at every point, so the winding contribution can be isolated within one framework without changing basis, solver or normalisation.